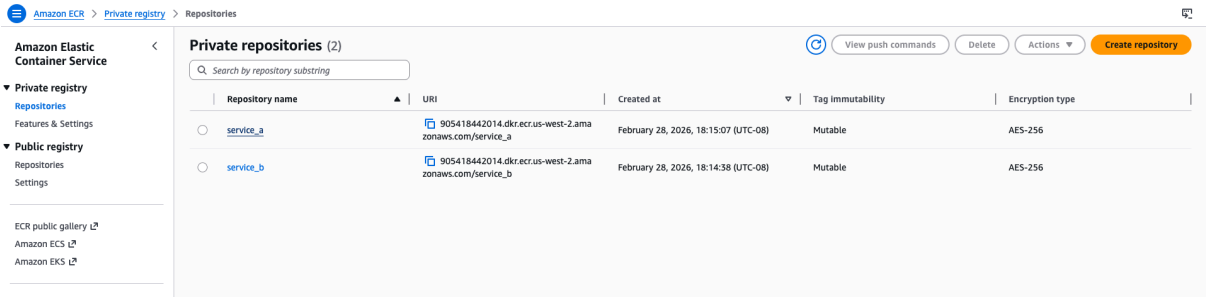
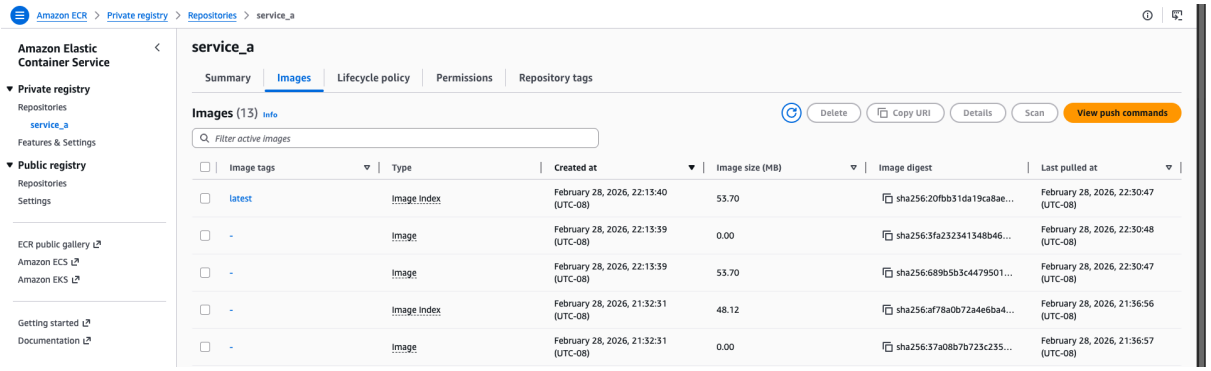
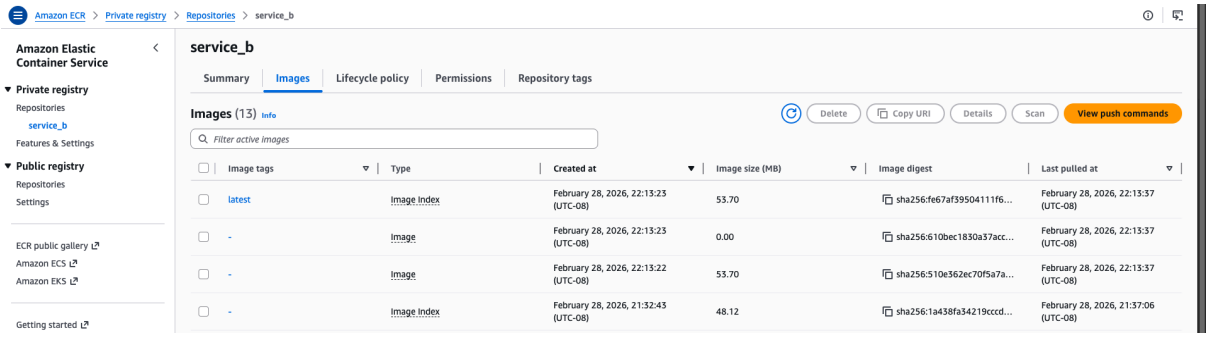
ECR







Instance 1

i-044ba7f97167a08ed (Web-Application-ServiceA)

PublicIPs: 35.87.35.150 PrivateIPs: 10.0.20.206

Instance 2

```
ubuntu@ip-10-0-20-246:~$ docker images
                                                                                                                                                                                  i Info →   U  In Use
IMAGE                                                                  ID             DISK USAGE   CONTENT SIZE   EXTRA
905418442014.dkr.ecr.us-west-2.amazonaws.com/service_b:latest          fe67af395041       220MB          53.7MB    U
ubuntu@ip-10-0-20-246:~$ docker ps
CONTAINER ID   IMAGE                                                   COMMAND                  CREATED        STATUS                    PORTS                                                   NAMES
3e34517f662c   905418442014.dkr.ecr.us-west-2.amazonaws.com/service_b:latest   "sh -c 'python /app/…"   8 minutes ago   Up 8 minutes (healthy)   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   app
ubuntu@ip-10-0-20-246:~$
```

i-03d3aea890bced210 (Web-Application-ServiceB)

PublicIPs: 44.255.19.125 PrivateIPs: 10.0.20.246

Health-Check + ALB DNS + `curl` responses

```
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_b/api/ping
{"message":"pong from service_b"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_a/api/ping
{"message":"pong from service_a"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_a/health
{"service":"service_a","status":"ok"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_a/api/ping
{"message":"pong from service_a"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_b/health
{"service":"service_b","status":"ok"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_b/api/ping
{"message":"pong from service_b"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://52.13.5.103:8080/health
{"service":"service_b","status":"ok"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops  curl http://44.251.219.75:8080/health
{"service":"service_a","status":"ok"}
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops
```

Scale Out - Service A

EC2 > Auto Scaling groups > webapp-asg-service-a

Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New

Images
AMIs
AMI Catalog

Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing
Load Balancers
Target Groups
Trust Stores

Auto Scaling
Auto Scaling Groups

Settings

Desired capacity	Scaling limits	Desired capacity type	Status
4	2 - 4	Units (number of instances)	-

Date created
Sat Feb 28 2026 21:26:38 GMT-0800 (Pacific Standard Time)

Details | Integrations | Automatic scaling | Instance management | Instance refresh | Activity | Monitoring | Tags

Activity notifications (0)

Actions ▼ | Create notification

Filter notifications

Send to	On instance action

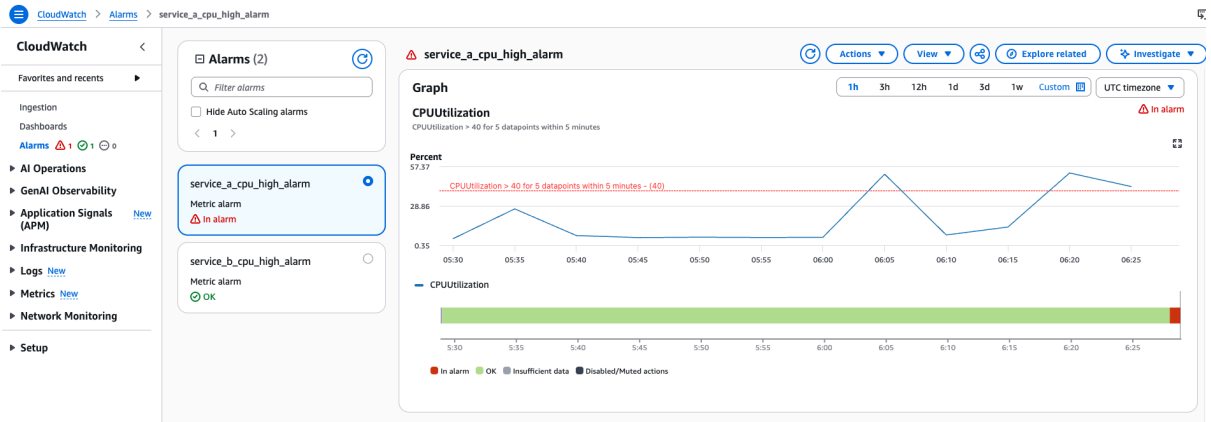
No notifications are currently specified

Create notification

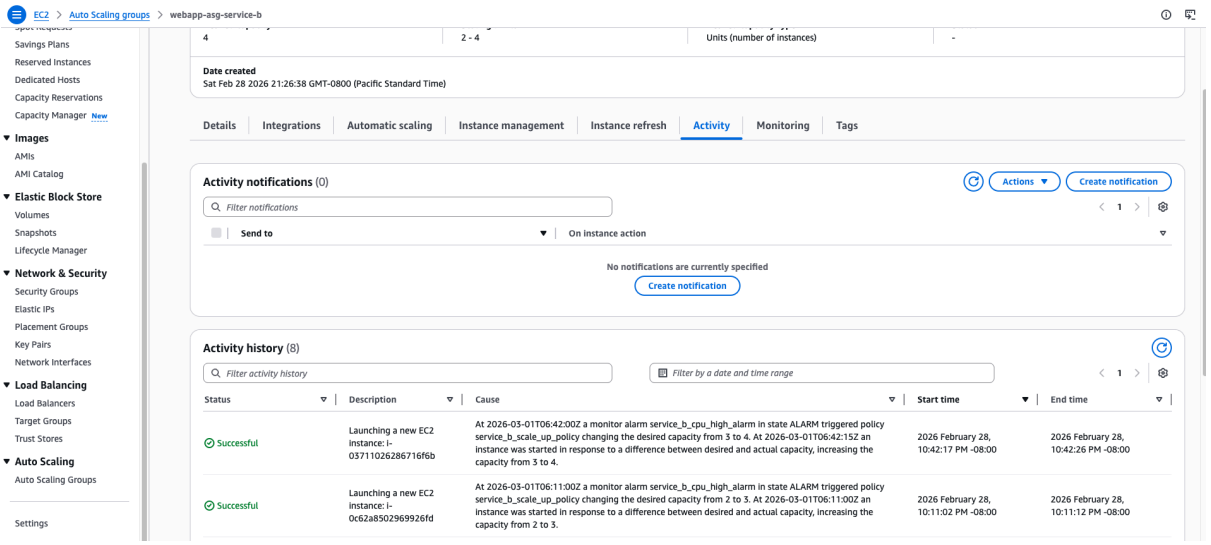
Activity history (8)

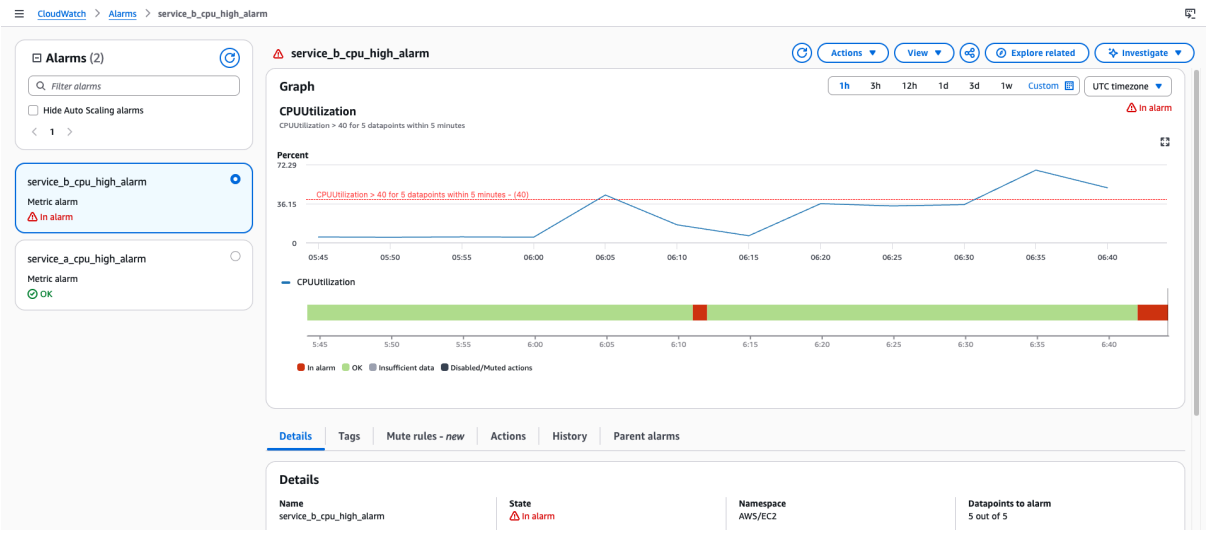
Filter activity history | Filter by a date and time range

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-005f29daa8b7db177	At 2026-03-01T06:30:01Z a monitor alarm service_a_cpu_high_alarm in state ALARM triggered policy service_a_scale_up_policy changing the desired capacity from 3 to 4. At 2026-03-01T06:30:04Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 3 to 4.	2026 February 28, 10:30:06 PM -08:00	2026 February 28, 10:30:16 PM -08:00
Successful	Launching a new EC2 instance: i-0efe8741d8e7e2573	At 2026-03-01T06:28:01Z a monitor alarm service_a_cpu_high_alarm in state ALARM triggered policy service_a_scale_up_policy changing the desired capacity from 2 to 3. At 2026-03-01T06:28:05Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 2 to 3.	2026 February 28, 10:28:07 PM -08:00	2026 February 28, 10:28:16 PM -08:00



Scale Out - Service B





Terraform Apply

```
iTerm2  Shell  Edit  View  Session  Scripts  Profiles  Window  Help                                   Sat Feb 28 9:26 PM
terraform apply -var-file="terraform.tfvars"
      + assign_ipv6_address_on_creation                = false
      + availability_zone                              = "us-west-2b"
      + availability_zone_id                           = (known after apply)
      + cidr_block                                     = "10.0.20.0/24"
      + enable_dns64                                   = false
      + enable_resource_name_dns_a_record_on_launch    = false
      + enable_resource_name_dns_aaaa_record_on_launch = false
      + id                                             = (known after apply)
      + ipv6_cidr_block_association_id                 = (known after apply)
      + ipv6_native                                    = false
      + map_public_ip_on_launch                        = true
      + owner_id                                       = (known after apply)
      + private_dns_hostname_type_on_launch            = (known after apply)
      + tags                                           = {
          + "Name" = "public-subnet-b"
        }
      + tags_all                                       = {
          + "Name" = "public-subnet-b"
        }
      + vpc_id                                         = (known after apply)
    }

  # module.network.aws_vpc.main will be created
  + resource "aws_vpc" "main" {
      + arn                                  = (known after apply)
      + cidr_block                           = "10.0.0.0/16"
      + default_network_acl_id               = (known after apply)
      + default_route_table_id               = (known after apply)
      + default_security_group_id            = (known after apply)
      + dhcp_options_id                      = (known after apply)
      + enable_dns_hostnames                 = true
      + enable_dns_support                   = true
      + enable_network_address_usage_metrics = (known after apply)
      + id                                   = (known after apply)
      + instance_tenancy                     = "default"
      + ipv6_association_id                  = (known after apply)
      + ipv6_cidr_block                      = (known after apply)
      + ipv6_cidr_block_network_border_group = (known after apply)
      + main_route_table_id                  = (known after apply)
      + owner_id                             = (known after apply)
      + tags                                 = {
          + "Name" = "main-vpc"
        }
      + tags_all                             = {
          + "Name" = "main-vpc"
        }
    }

Plan: 31 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value:
```

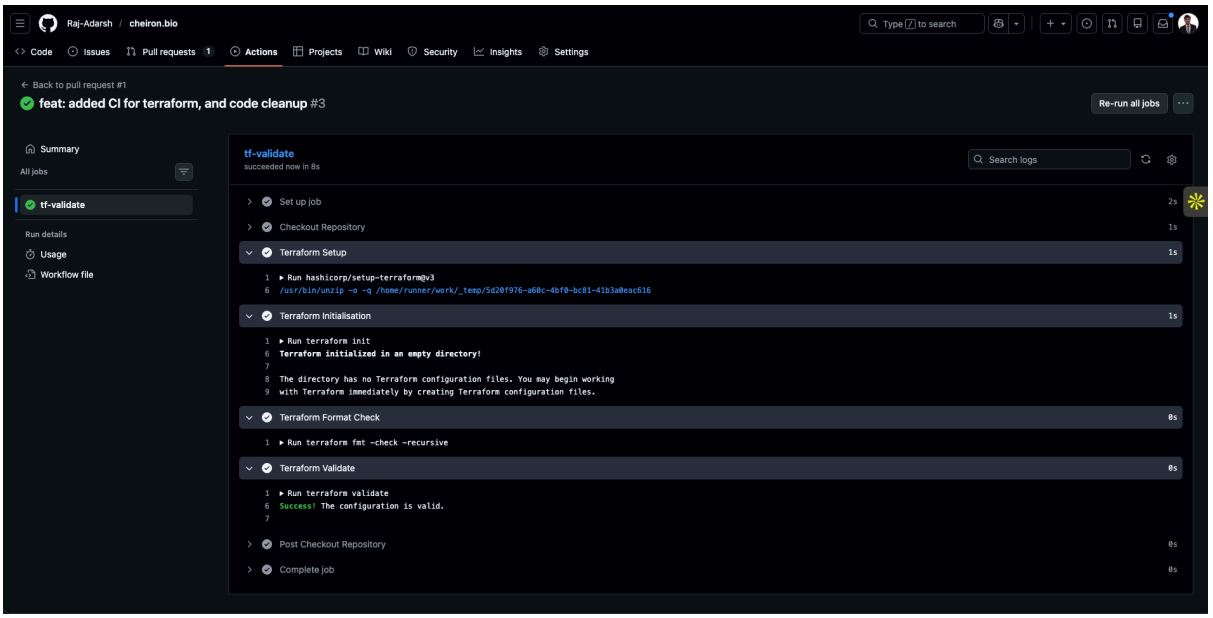
Terraform Destroy

```
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 4m50s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 4m50s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 4m50s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 5m0s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 5m0s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 5m0s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 5m10s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 5m10s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 5m10s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 5m20s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 5m20s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 5m20s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 5m30s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 5m30s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 5m30s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 5m40s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 5m40s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 5m40s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 5m50s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 5m50s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 5m50s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 6m0s elapsed]
module.network.aws_internet_gateway.igw: Still destroying... [id=igw-0428dfe1139e26f6a, 6m0s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 6m0s elapsed]
module.network.aws_internet_gateway.igw: Destruction complete after 6m9s
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 6m10s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 6m10s elapsed]
module.ec2.aws_autoscaling_group.service_b_asg: Still destroying... [id=webapp-asg-service-b, 6m20s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Still destroying... [id=webapp-asg-service-a, 6m20s elapsed]
module.ec2.aws_autoscaling_group.service_a_asg: Destruction complete after 6m27s
module.alb.aws_lb_target_group.service_a_tg: Destroying... [id=arn:aws:elasticloadbalancing:us-west-2:905418442014:targetgroup/service-a-tg/c10e377ed3a1477a]
module.ec2.aws_launch_template.service_a_lt: Destroying... [id=lt-0d3ae21cec3d4fa4d]
module.ec2.aws_autoscaling_group.service_b_asg: Destruction complete after 6m27s
module.network.aws_subnet.public_subnet_b: Destroying... [id=subnet-0d32f1d4c5c86b237]
module.network.aws_subnet.public_subnet_a: Destroying... [id=subnet-0b01067569c8c28ae]
module.alb.aws_lb_target_group.service_b_tg: Destroying... [id=arn:aws:elasticloadbalancing:us-west-2:905418442014:targetgroup/service-b-tg/6e85f4be812bf67f]
module.ec2.aws_launch_template.service_b_lt: Destroying... [id=lt-0008d3dee5129e140]
module.alb.aws_lb_target_group.service_b_tg: Destruction complete after 1s
module.alb.aws_lb_target_group.service_a_tg: Destruction complete after 1s
module.ec2.aws_launch_template.service_b_lt: Destruction complete after 1s
module.ec2.aws_launch_template.service_a_lt: Destruction complete after 1s
module.ec2.aws_iam_instance_profile.ec2_profile: Destroying... [id=my-instance-profile]
module.network.aws_security_group.app_sg: Destroying... [id=sg-0b6dea77fc5f8dd43]
module.network.aws_subnet.public_subnet_b: Destruction complete after 1s
module.network.aws_subnet.public_subnet_a: Destruction complete after 1s
module.ec2.aws_iam_instance_profile.ec2_profile: Destruction complete after 1s
module.ec2.aws_iam_role.ec2_role: Destroying... [id=my-role]
module.network.aws_security_group.app_sg: Destruction complete after 1s
module.network.aws_vpc.main: Destroying... [id=vpc-0d0859f858c96ced7]
module.ec2.aws_iam_role.ec2_role: Destruction complete after 0s
module.network.aws_vpc.main: Destruction complete after 0s

Destroy complete! Resources: 31 destroyed.
 AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/infra  feat/devops
```

Extras

Verify Endpoint

```
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/services  feat/devops ±  ./verify_endpoints.sh app-lb-470532594.us-west-2.elb.amazonaws.com
[RESPONSE] http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_a/health => {"service":"service_a","status":"ok"}
[RESPONSE] http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_a/api/ping => {"message":"pong from service_a"}
[RESPONSE] http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_b/health => {"service":"service_b","status":"ok"}
[RESPONSE] http://app-lb-470532594.us-west-2.elb.amazonaws.com/service_b/api/ping => {"message":"pong from service_b"}
[PASS] All checks passed for service_a and service_b on app-lb-470532594.us-west-2.elb.amazonaws.com
AWS: infra  adarshraj@Adarshs-MacBook-Pro  ~/Documents/devops/services  feat/devops ±
```

Github Actions